

Williams Racing Performance Analysis (2015-2019)

How can past results inform current strategy at Williams Racing?

Presented by: Frank Dong

Last Updated: September 28th, 2025

Table of Contents

- Purpose Statement
- Questions, KPIs & Hypotheses
- 1 – Grid-to-Finish Performance
- 2 – Qualifying Sector Deficits
- 3 – Driver Consistency
- Recommendations and Q&A

Purpose Statement

Outlining the objectives that will shape Williams' future recommendations

Objectives

- **Analyse** Williams' 2015-2019 performance across domains of race weekend execution:
 - Race performance
 - Qualifying sectors
 - Driver consistency
- **Surface** insights informing future strategic decisions
- **Recommend** development priorities, whether in driver development, technical packages, or racecraft execution

Questions, KPIs, & Hypotheses

How do we link race strategy to data analysis?

Connecting Strategy to Analysis

#	Strategic Question	KPI	Hypothesis
1	Where and why did Williams lose the most race positions?	Avg. race positions gained/lost	More positions were lost at technical tracks compared to midfield rival average
2	Time lost in qualifying: technical or power sectors? *	Sector time gap to fastest midfield rival	More time was lost in technical sectors
3	Were rookies less consistent than experienced teammates?	Lap time standard deviation, per race	Rookies had more variable lap times

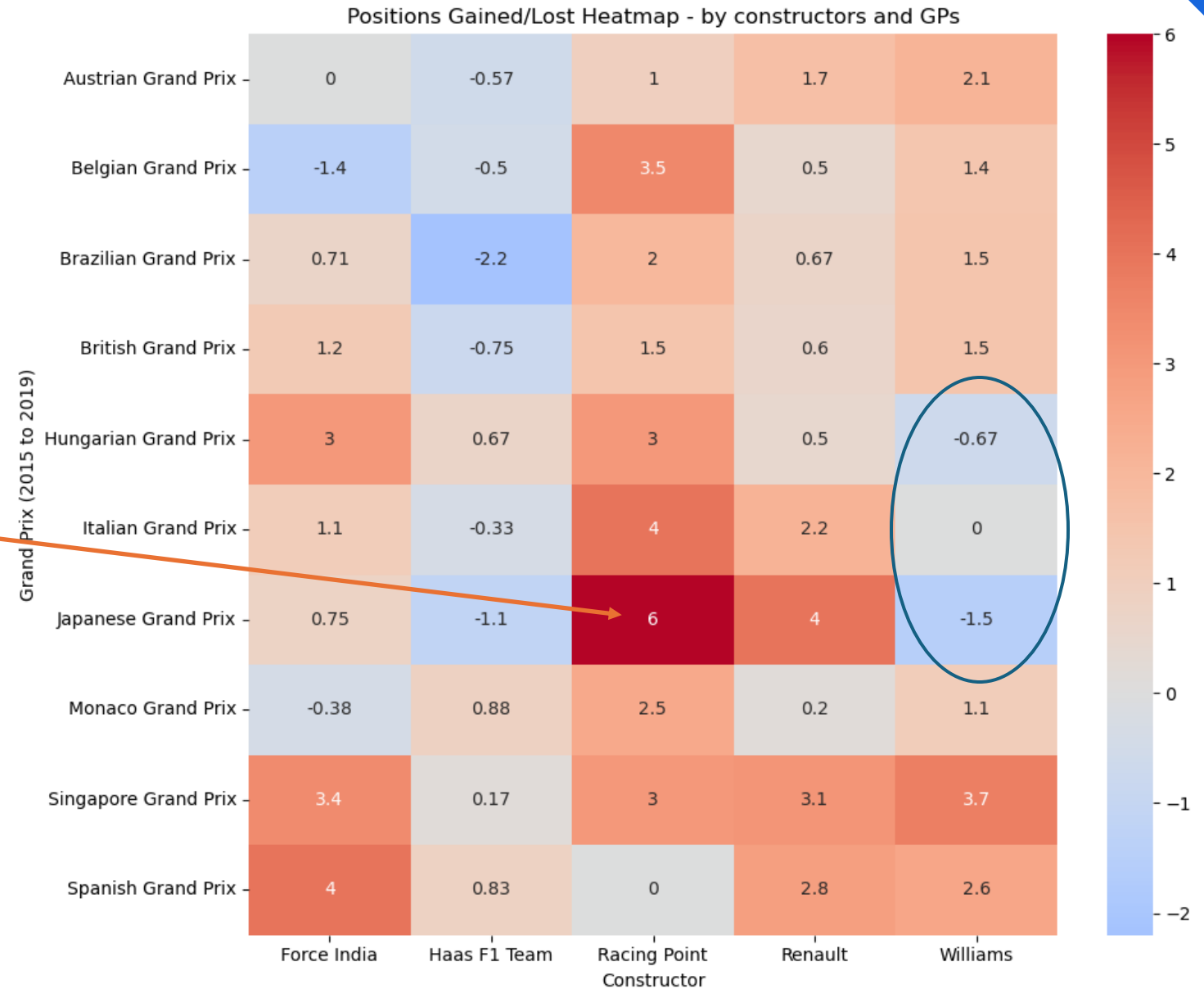
* Limited to 2018 and 2019 race data

1 - Grid-to-Finish Performance

Where and why did Williams lose the most race positions?

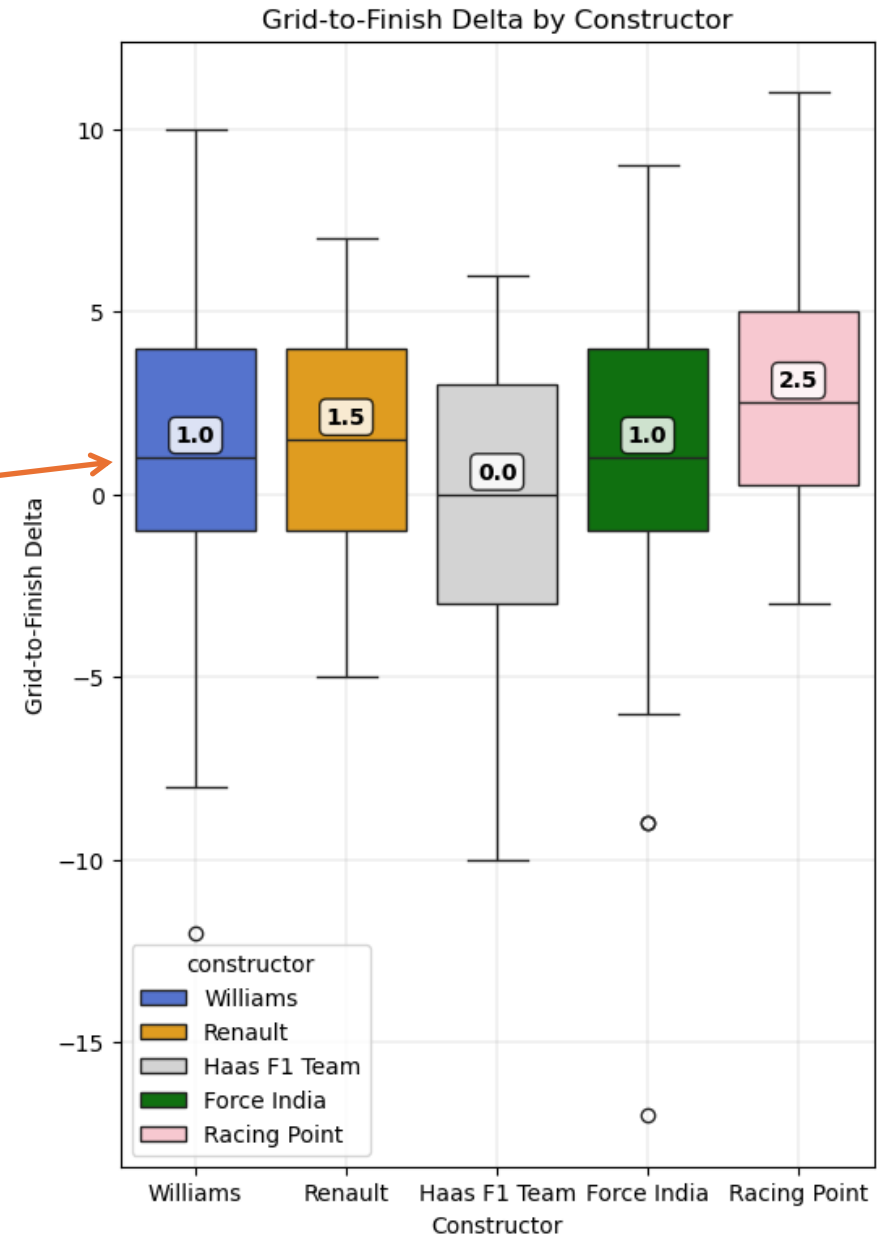
1 – Grid-to-Finish Performance

- **Data covers** 10 races, 5 teams, 2015-2019
- **Heatmap shows** positions gained or lost per team & race
- **Racing Point:** gained most positions, esp. at the Japanese GP
- **Williams' trend:** typically, lost places at technical tracks (Japanese, Hungarian)
- **Key Question:** Are technical tracks a challenge for Williams?



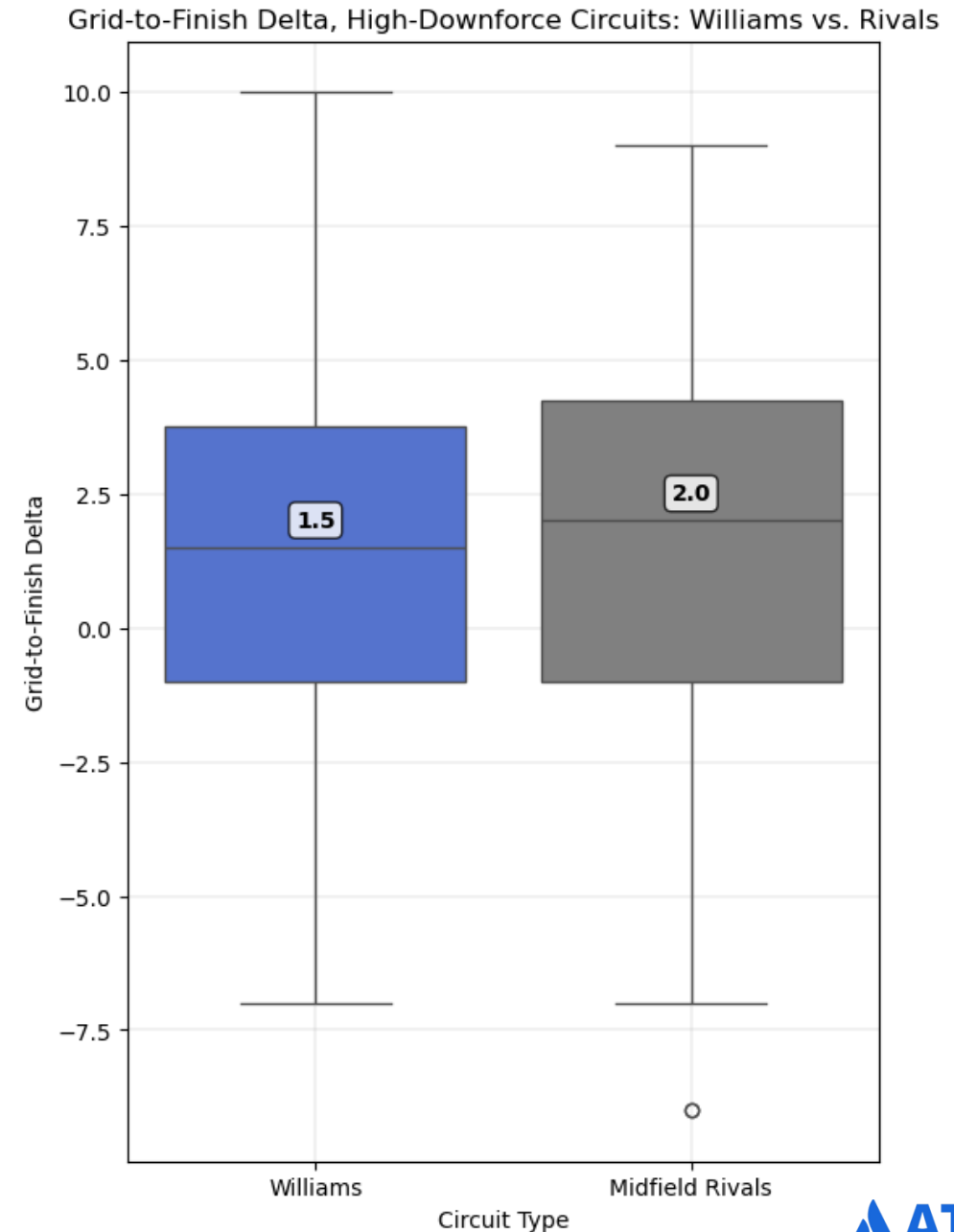
1 – Grid-to-Finish Performance

- Boxplots summarise 10 GPs over 5 seasons for each midfield team
- Williams' median finish = positions gained
- Results are **highly variable** for Williams – big swings in gains and losses.



1 – Grid-to-Finish Performance

- Grey boxplots indicate average results of midfield rivals (grey):
 - Haas
 - Renault
 - Force India / Racing Point
- Race sessions are grouped by circuit type: power, balanced, and technical
- Technical circuits: smaller box = Williams turns out to be more consistent.



1 – Hypothesis Testing

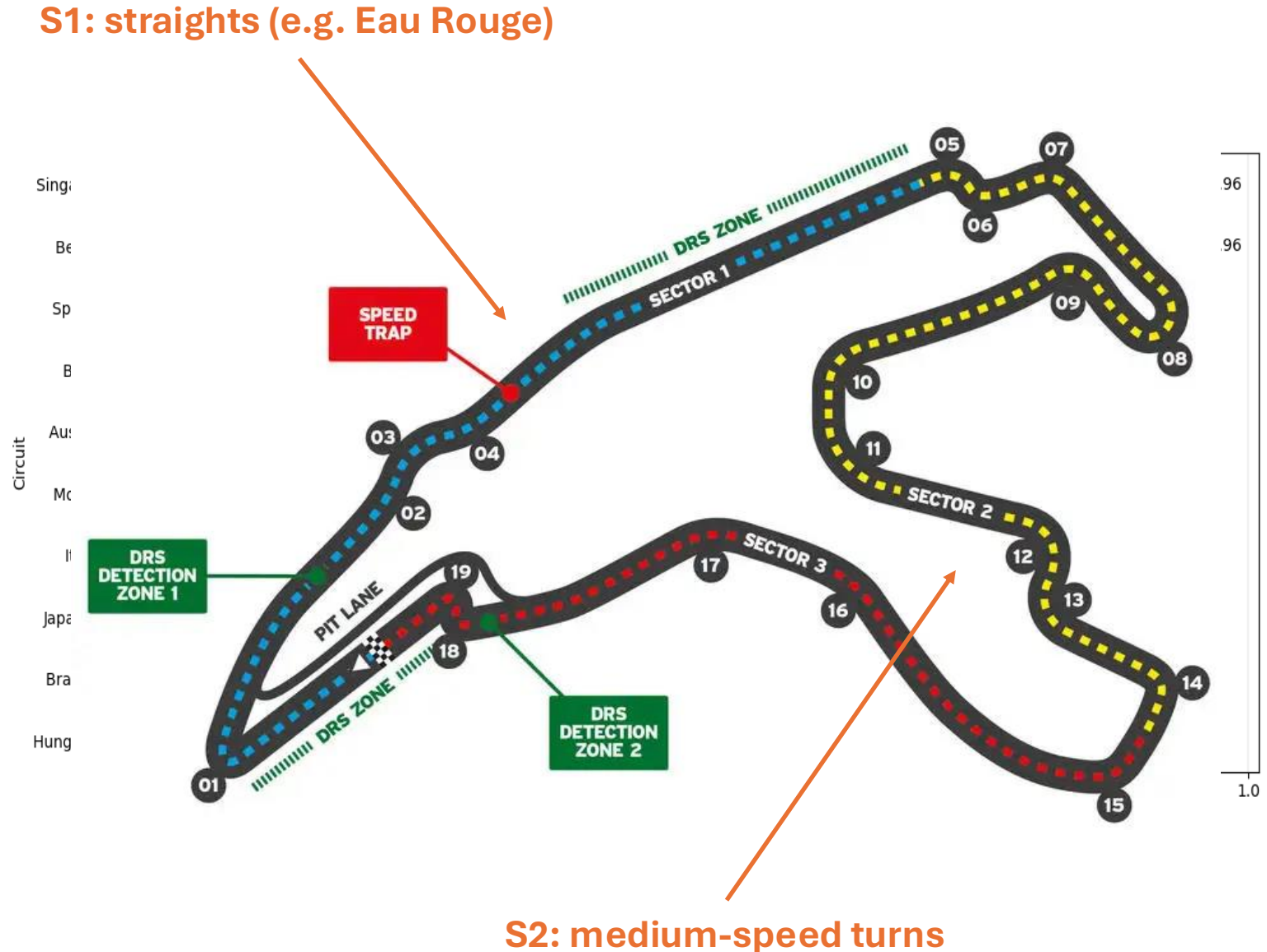
- Tested if Williams **lost more race positions than rivals at technical circuits.**
- **Welch's t-test** and **Mann-Whitney U** to validate results.
- **No significant difference** in position loss at technical circuits.
- Next: Examine qualifying sector time gaps
 - Where does Williams lose time, in technical or power sectors?

2 - Qualifying Sector Deficits

Did Williams lose time in qualifying in the technical or power sectors?

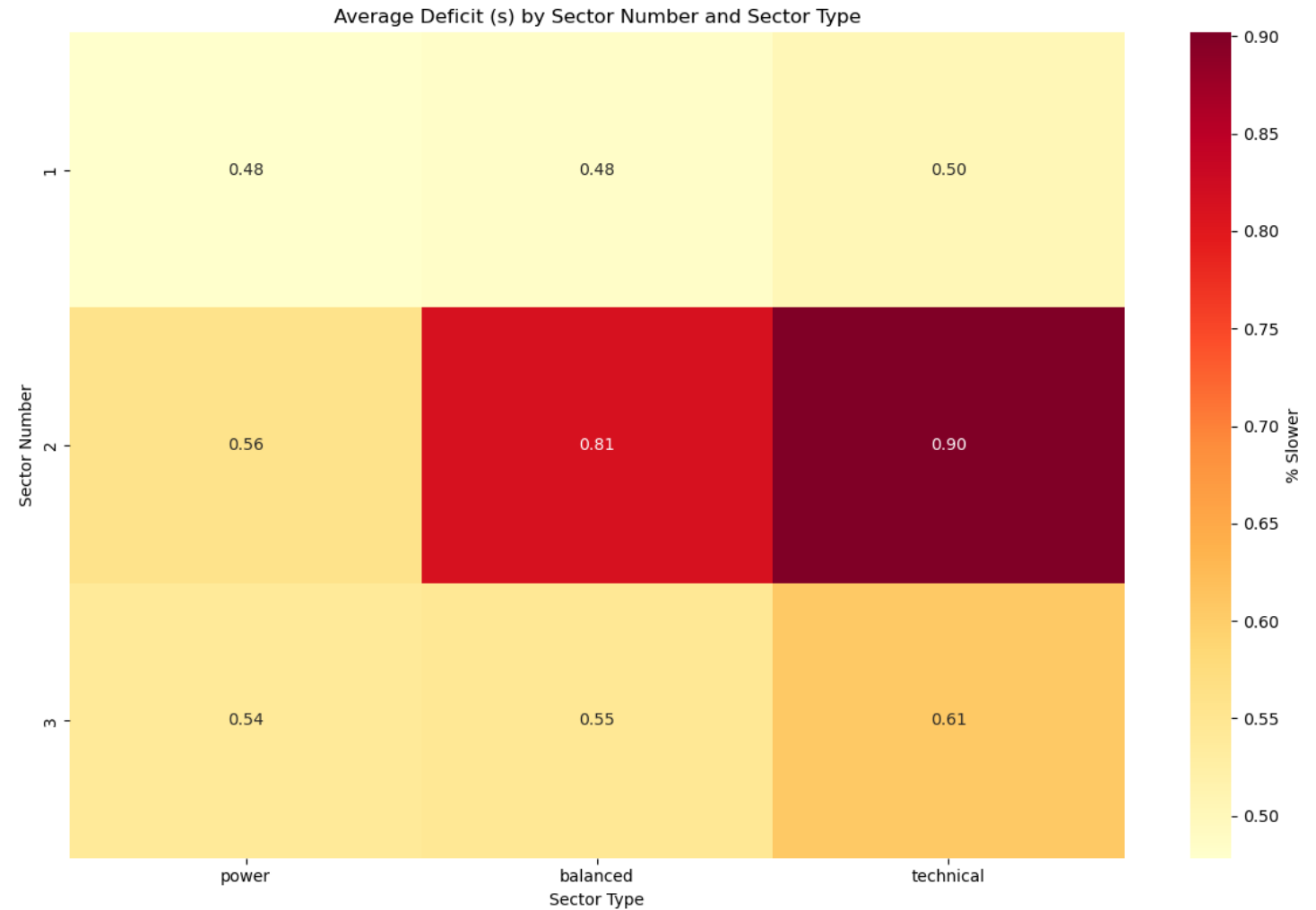
2 – Sector Deficits

- Analysis uses **qualifying data from 2018–2019** (Fast-F1 API limitation)
- Bars indicate **average time lost by Williams** to fastest midfield rival in each Grand Prix
- No clean pattern at the circuit-level. Deficits vary by circuit type.
- To diagnose strategy, we break down results by circuit sector



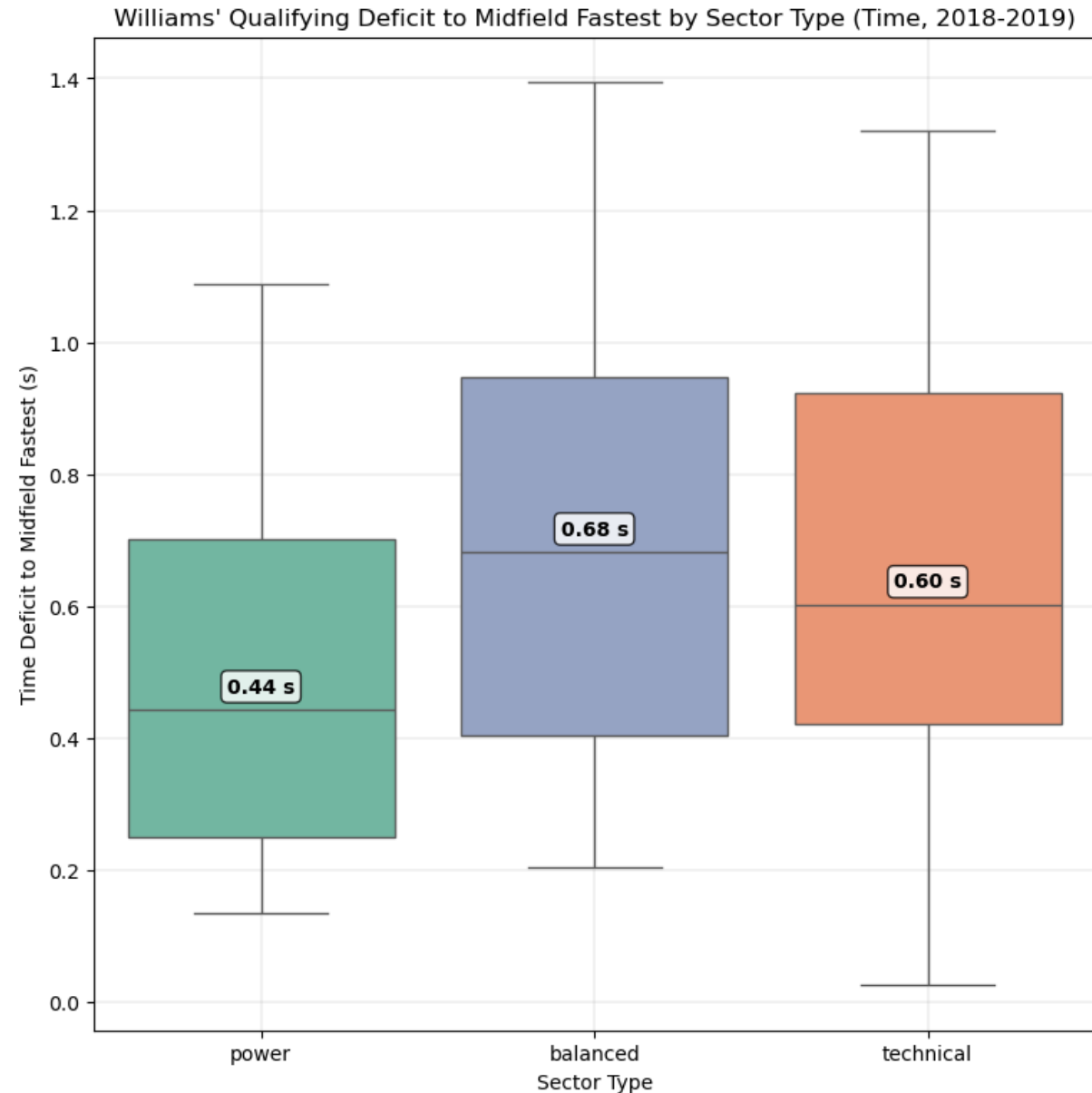
2 – Sector Deficits

- Heatmap shows where Williams loses the most time by sector type and number
- Largest deficits in **Technical Sector 2** – 0.9 s lost on average.
- **Balanced S2** and **Technical S3** also stand out for big qualifying gaps.
- Pattern suggests that technical sectors challenge William more than power sectors.



2 – Sector Deficits

- **Smallest and most consistent** qualifying deficits in **power sectors** (0.44 s).
- **Technical sectors** – **higher typical deficit** (0.6 s) and 2nd-widest variability.
- Surprisingly, **balanced sectors** show the greatest spread and avg deficit (0.68 s)



2 – Hypothesis Testing

- Tested if Williams **loses more time in technical sectors compared to power sectors**
- **Mann-Whitney U** to validate results (small, non-normal samples).
- **No significant evidence** that Williams had a larger deficit in technical sectors.
- Large variability was observed in balanced and technical sectors.
- Next: Examine role of driver experience in lap time consistency
 - Do rookies have less consistent lap times?

3 – Driver Consistency

Did rookies have more variable lap times?

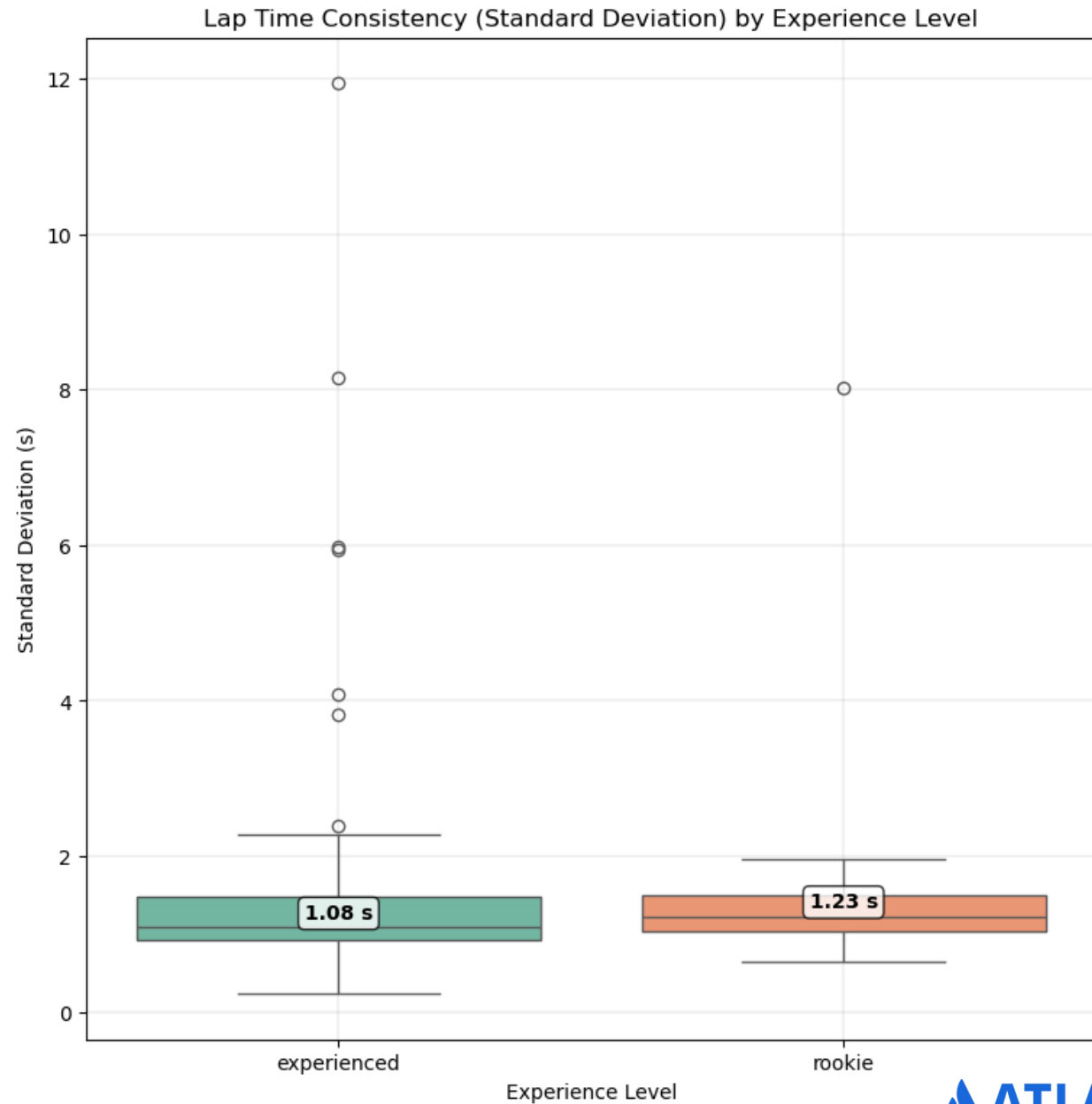
3 – Driver Consistency

- Lap time consistency = standard deviation of lap times in one race
- Massa, Bottas = over-indexed
- Kubica, Sirotkin, Russell = under-indexed
- Can also identify:
 - **clusters** (typical consistencies)
 - **outliers** (rare performances or race incidents)



3 – Driver Consistency

- Rookies are **less consistent** compared to experienced drivers.
- Lower median std. dev. = more consistent lap times
- Outliers in both groups' results are more clearly visible.



3 – Hypothesis Testing

- Tested if Williams' **rookies had more variable lap times** compared to experienced drivers.
- **Mann-Whitney U** test used to validate results – due to inclusion of outliers.
- Statistically, rookies have **greater** lap time variance than veterans.
- Findings show that in-race consistency develops with experience.

Conclusions, Recommendations, Q&A

Key takeaways, actionable next steps, and your questions.

Conclusions

1. Williams showed the greatest volatility in race position changes compared to midfield rivals.
2. Balanced sectors, not technical ones, revealed the largest qualifying gaps.
3. Rookies were statistically less consistent than experienced teammates, confirming higher lap-to-lap variance for inexperienced drivers.

Recommendations

1. Implement a live sector-tagging and delta tracking system for use during race sessions
2. Target stability in balanced circuits/sectors by trialling aero-mechanical setup ranges
3. Prioritise consistency training over raw pace development for rookie drivers through simulator work and tailored coaching.

Thank you for listening!

Any questions?